

Lecture 19

m-to-1.

Last time: bijection, sum, product, division.

$\{\text{objects}\} \longleftrightarrow \{\text{ways of making choices}\}$

3 basic counts:
(no repetitions)

$$n! , P(n, r) = \frac{n!}{(n-r)!} , \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

ex: Sheep. Observe 108 legs.

$$\# \text{ sheep} = \frac{\# \text{ legs}}{\# \text{ legs per sheep}} = \frac{108}{4} = 27.$$

I Counting and Algebra

II Perm & comb with repetitions

x, y variables.

$$\begin{aligned} (x+y)^2 &= (x+y)(x+y) = xx + xy + yx + yy \\ &= x^2 + 2xy + y^2 \end{aligned}$$

4 original terms
↓
3 terms.

$$(x+y)^3 = (x+y)(x+y)(x+y)$$

$$= xxx + xxy + xyx + xyy + yxx + yxy + yyx + yyy$$

Strings of x, y
of length 3.

$$= x^3 + \binom{3}{1} x^2 y + \binom{3}{2} x y^2 + y^3$$

group by
of x and y .

Binomial Thm: If $n \geq 1$ then

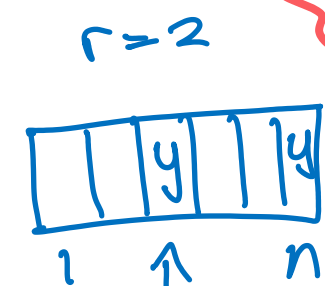
$$(x+y)^n = \binom{n}{n} x^n + \binom{n}{n-1} x^{n-1} y + \dots + \binom{n}{1} x y^{n-1} + \binom{n}{0} y^n.$$

$$= \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$$

Proof:

Assume $n \geq 1$. Observe that the expansion of $(x+y)^n$ consists of 2^n terms corresponding to all xy -strings of length n , since

each term arises by choosing the variable x or y from each parenthesized factor of $(x+y)(x+y)\dots(x+y)$.



$S \subseteq [n]$
 $|S|=2$

Grouping monomials, for $r \in \{0, \dots, n\}$ the coefficient of $x^{n-r}y^r$ is the number of xy -strings of length n with exactly r y 's, which is equal to the number of r -subsets of $\{1, \dots, n\}$, which is $\binom{n}{r}$.

bijection

Thus

$$(x+y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r. \quad \square$$

Notation: $[n] = \{1, \dots, n\}$. $\binom{n}{r} = 0$ if $r > n$.
 $0! = 1$.

Combinatorial Identities:

Prop 1: If $n \geq 1$ then $\boxed{\binom{n}{r} = \binom{n}{n-r}}$.

Proof #1 (Algebra): $\binom{n}{r} = \frac{n!}{(n-r)! r!}$
 $= \frac{n!}{(n-r)! (n-(n-r))!}$
 $= \binom{n}{n-r}.$

Proof #2 (Bijection) Let $n \geq 1$, $0 \leq r \leq n$. Define

$$A = \{S \subseteq [n] : |S| = r\}, \quad |A| = \binom{n}{r}.$$

$$B = \{S \subseteq [n] : |S| = n-r\}, \quad |B| = \binom{n}{n-r}.$$

Observe that $f: A \rightarrow B$ by

$$f(S) = \{1, \dots, n\} - S.$$

is a bijection since

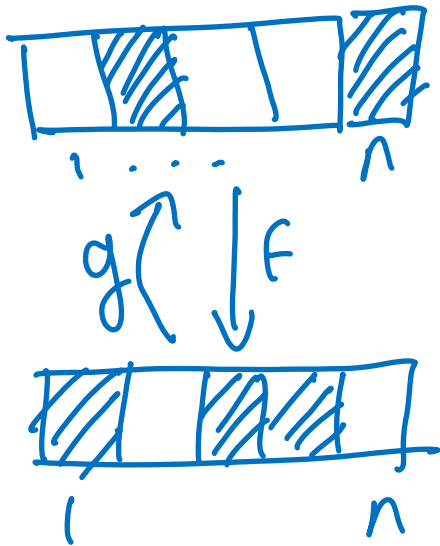
$$g: B \rightarrow A \text{ by } g(T) = \{1, \dots, n\} - T$$

is its inverse.

Thus,

$$\binom{n}{r} = |A| \stackrel{\text{bijection}}{=} |B| = \binom{n}{n-r}.$$

□



Prop 2: If $n \geq 1$ then $2^n = \binom{n}{n} + \binom{n}{n-1} + \binom{n}{n-2} + \dots + \binom{n}{1} + \binom{n}{0}$

$$2^n = \sum_{r=0}^n \binom{n}{r}$$

$\mathcal{P}([n])$

Proof #1 (Bijection): We will count the subsets of $[n]$ in two different ways:

Method 1: A subset $S \subseteq [n]$ is uniquely specified by the following choices:

	<u># ways</u>
$1 \in S?$	2
$2 \in S?$	2
$\vdots$	
$n \in S?$	2

By the product rule, $|\mathcal{P}([n])| = 2^n$.

Method 2: A subset $S \subseteq [n]$ is uniquely specified
by the following choices:

(1) Choose $0 \leq r \leq n$.

ways $\downarrow$
 $n+1$.

choice tree



(2) Given r , choose $S \subseteq [n]$
s.t. $|S| = r$.

$\binom{n}{r}$
 $\downarrow$
depend on
(1)

By the sum rule applied to

$$\mathcal{P}([n]) = A_0 \cup A_1 \cup \dots \cup A_n$$

where $A_r = \{S \subseteq [n] : |S| = r\}$,

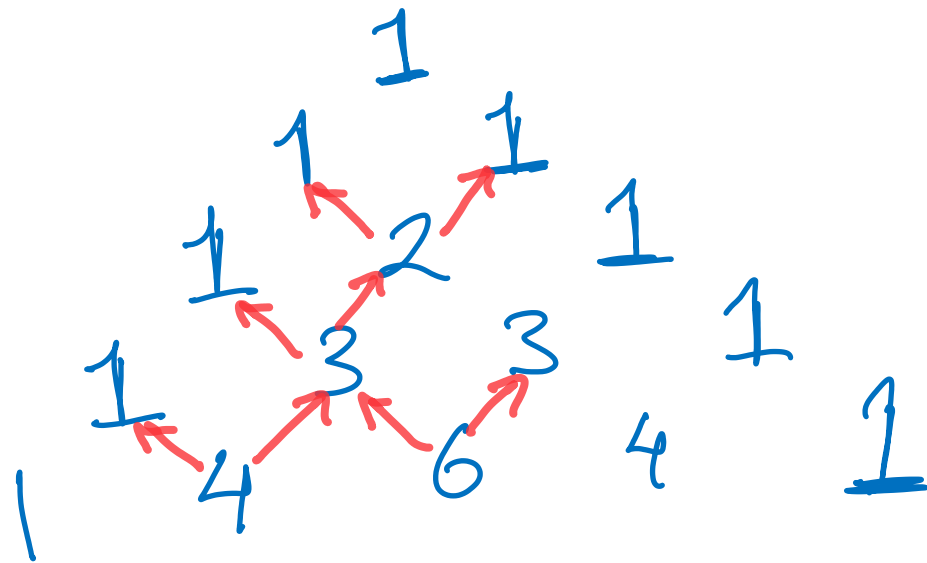
$$|\mathcal{P}([n])| = \sum_{r=0}^n \binom{n}{r}.$$

Thus, $2^n = \sum_{r=0}^n \binom{n}{r}$, as desired. $\square$

Creativity: What are the two sides of the identity counting?

Prop 3: (Pascal's Triangle) If $n \geq 1$ and $1 \leq r \leq n$ then

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$$



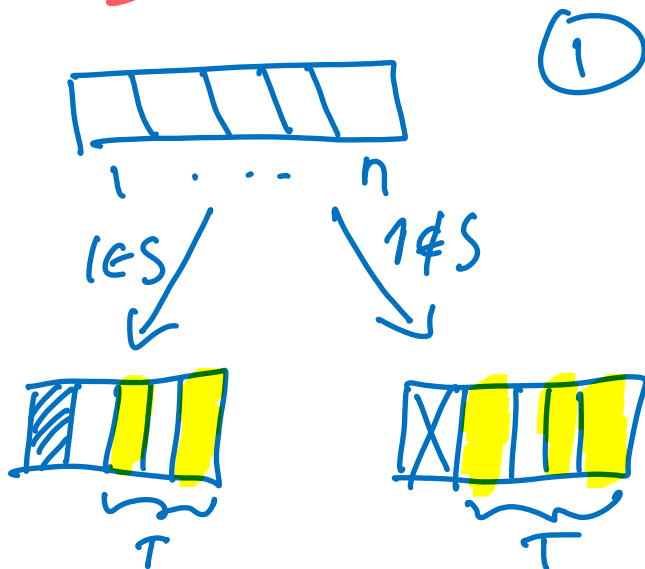
Proof: We will count the number of r -subsets of $[n]$ in two different ways.

Method 1: By defn, this is $\boxed{\binom{n}{r}}$.

Method 2: An r -subset $S \subseteq [n]$ is uniquely ^{bijection} specified by the following process:

$n=5$
 $r=3$

#ways



① Decide if $1 \in S$.

② If $1 \in S$, choose an $(r-1)$ -subset $T \subseteq \{2, \dots, n\}$ and let $S = \{1\} \cup T$.

③ If $1 \notin S$, choose an r -subset $T \subseteq \{2, \dots, n\}$ and let $S = T$.

By the sum rule, the total # of r -subsets is $\boxed{\binom{n-1}{r-1} + \binom{n-1}{r}}$.

Thus, $\boxed{\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}}$

easy to confuse

Warning: n and r

II Counting w/ repetitions

So far:

		repetitions allowed?	
		Y	N
order matters?	Y	n^r	$P(n, r)$
	N	$\binom{r+n-1}{n-1}$	$\binom{n}{r}$

of arrangements
of r objects
chosen from
 n distinct
objects.

ex: # of 7-digit phone numbers with digits in $\{0, \dots, 9\}$.

$r = 7$
 $n = 10$

$$|\{0, \dots, 9\} \times \dots \times \{0, \dots, 9\}| = |\{0, \dots, 9\}|^7$$

7 times

$$= 10^7.$$

(Cartesian product).

ex 2: How many ice cream sundaes of of
 2 scoops can you make from
 $\{ \underset{V}{\text{Vanilla}}, \underset{C}{\text{Chocolate}}, \underset{M}{\text{Mint}} \}$.
 order doesn't matter.
 + reps allowed.

$\begin{matrix} VV \\ CC \\ MM \end{matrix}$
 $\begin{matrix} VC \\ CM \\ MV \end{matrix}$
 ~~$\begin{matrix} VC \\ MC \\ VM \end{matrix}$~~
 $\left. \vphantom{\begin{matrix} VV \\ CC \\ MM \end{matrix}} \right\} 6.$

3 scoops:

$\begin{matrix} MMM \\ CCC \\ VVV \end{matrix}$
 $\begin{matrix} MMV \\ MMC \\ CCV \\ CCM \\ VVM \\ VVC \end{matrix}$
 MVC
 $\left. \vphantom{\begin{matrix} MMM \\ CCC \\ VVV \end{matrix}} \right\} \underline{\underline{10}}.$

5 scoops?

$r = 5, n = 3.$

Process: Create one unlabeled token per scoop

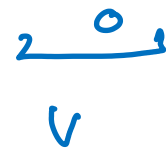
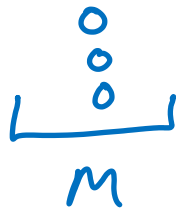
o o o o o .

A sundae is uniquely specified by a placement of 5 tokens into 3 boxes labeled M, C, V.

ex:

3M
1C
1V

≡



#sundaes = # ways of putting 5 tokens in 3 labeled boxes.

Sneaky :



$$l=3$$

$$r=5$$

||| bijection.



||| bijection.

0 0 0 1 0 1 0

$$21 = \frac{7 \cdot 6}{2} = \binom{7}{2}.$$

7-bit string
w/ 2 1's

0 : token

1 : partitions.

Theorem: The number of arrangements of $\boxed{r}$ objects chosen from $\boxed{n}$ objects where order doesn't matter & repetitions are allowed is $\boxed{\binom{r+n-1}{n-1}}$. ~ "categories of objects"

Proof: Such an arrangement is uniquely specified by a placement of r unlabeled tokens into n labeled boxes, which corresponds to a binary string with r 0's (corresponding to the tokens) and $n-1$ 1's (corresponding to the dividers). The number of such strings is $\binom{r+n-1}{n-1}$. $\square$